

A Theory of Fads, Fashion, Custom, and Structural Change as Informational Cascades

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Hirshleifer, Ivo Welch, 1992

Studied and replicated by: Jamal Verdiyev

“An informational cascade occurs when it is optimal for an individual, having observed the actions of those ahead of him, to follow the behavior of the preceding individual without regard to his own information. We argue that localized conformity of behavior and the fragility of mass behaviors can be explained by informational cascades.”

Let them alone: they be blind leaders of the blind. And if the blind lead the blind, both shall fall into the ditch.
[Matthew 15:14]

01

Motivation & Big picture



Motivation: Why Uniformity Happens

Why Do People Do the Same Thing?

- **We observe conformity + sudden reversals**
- **Four mechanisms:**
 1. sanctions
 2. taste for conformity
 3. payoff externalities
 4. communication
- This paper: **uniformity from information**

Core Idea: Informational Cascade

- People see **actions**, not signals
- A cascade: actions become **independent of private signals**
- Rational imitation → collective mistakes possible

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The Model

Basic Model (Sequential Social Learning)

- Hidden state $V \in \{0,1\}$, prior $P(V = 1) = 0.5$
- Each agent gets a private signal $s_i \in \{H, L\}$ with accuracy $p > 0.5$, where $p = P(s_i = V)$
- Chooses action $a_i \in \{Adopt, Reject\}$ to maximize expected payoff
- Observes past actions (PAO)

How a Cascade Starts

- After a few consistent actions, one signal can't overturn the posterior
- Rule of thumb: “two same early actions” $\rightarrow$ others follow
- After cascade starts: **new actions add little information**

PAO

- Private signals are **hidden**
- Only **actions** are observed
- Actions are a **compressed statistic**



cascades are **sticky** and can be **wrong**

PSO

- Private signals are **observed**
- Information **aggregates** over time
- Cascades are **weaker/rarer**



Learning is **more accurate**

03

Key Mechanisms & Predictions

Fragility & Public Signals

Fragility: Small News Can Break Big Herds

- Cascades are “thin”: repetition $\neq$ more evidence
- Rare public signals can **break / redirect** cascades



“Even rare public signals improve correctness”

- Leads to testable predictions (see Table 2 for numbers)

Authors' Results (Theory)

1. Existence

Cascades arise with high probability (under mild conditions).

2. Inefficiency

Cascades can be wrong: learning stops once actions stop revealing new info.

3. Fragility

Cascades are fragile: small public information can overturn a long cascade.

Authors' Quantitative Predictions

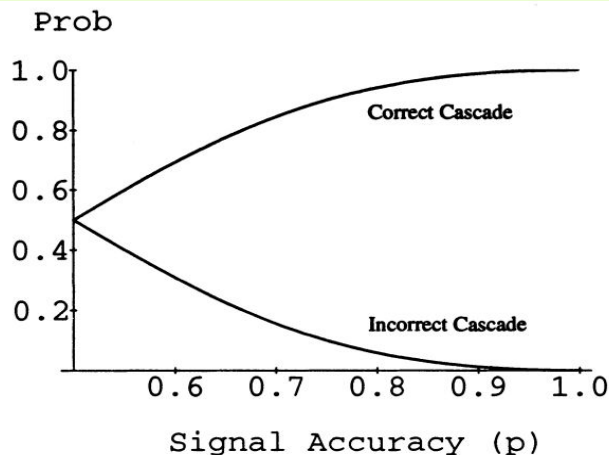


FIG. 1.—Probability of a correct and an incorrect cascade as a function of p (p is the probability that the signal is high [H] given that the true value is high [eq. (1)]). Even for large p , the probability of ending up in the wrong cascade is considerable.

“Wrong cascades remain possible even when signals are fairly accurate.”

Authors' Quantitative Predictions

TABLE 2
EFFECT OF MULTIPLE INFORMATION RELEASES

p	NO PUBLIC INFORMATION			PUBLIC INFORMATION (1)			PUBLIC INFORMATION (10)		
	Cascade		Expectation of Difference (3)	Cascade		Expectation of Difference (6)	Cascade		Expectation of Difference (9)
	Up (1)	Down (2)		Up (4)	Down (5)		Up (7)	Down (8)	
.55	.5645	.4355	.262	.5857	.4132	.451	.6767	.3200	1.691
.65	.6950	.3050	.783	.7396	.2601	1.236	.8986	.1005	4.577
.75	.8100	.1900	1.242	.8633	.1366	1.950	.9826	.0172	6.874
.85	.9065	.0935	1.623	.9480	.0516	2.494	.9989	.0011	8.940
.95	.9724	.0276	1.890	.9884	.0116	2.858	1.0000	.0000	10.982

Authors' Quantitative Predictions

Result 4. In the numerical example on fads the probability of a cascade reversal is greater than .0935, compared to a probability of only .05 that the correct choice changes. Therefore, the probability of convention changes can be substantially higher when only previous actions are observed than under full information.

“agents discount past information because fundamentals may have changed.”

04+

Phenomena

Examples (Where It Shows Up)

Intuition in the Real World

- Politics (polls, primaries)
- Medicine (treatment “fads”)
- Finance (runs / IPO demand / bids)
- Peer effects & stigma

Fads (Why Behavior Switches)

Sudden Shifts in Collective Behavior

- **Fad:** a rapid, widespread switch in actions (adopt → reject)
without a comparable change in fundamentals
- **Why it happens:** after a cascade starts, **actions stop revealing new information**
- **So:** small public news (or a small change fundamentals changed) can **flip** the crowd

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Replication & Extra Outcomes

My Replication (Setup)

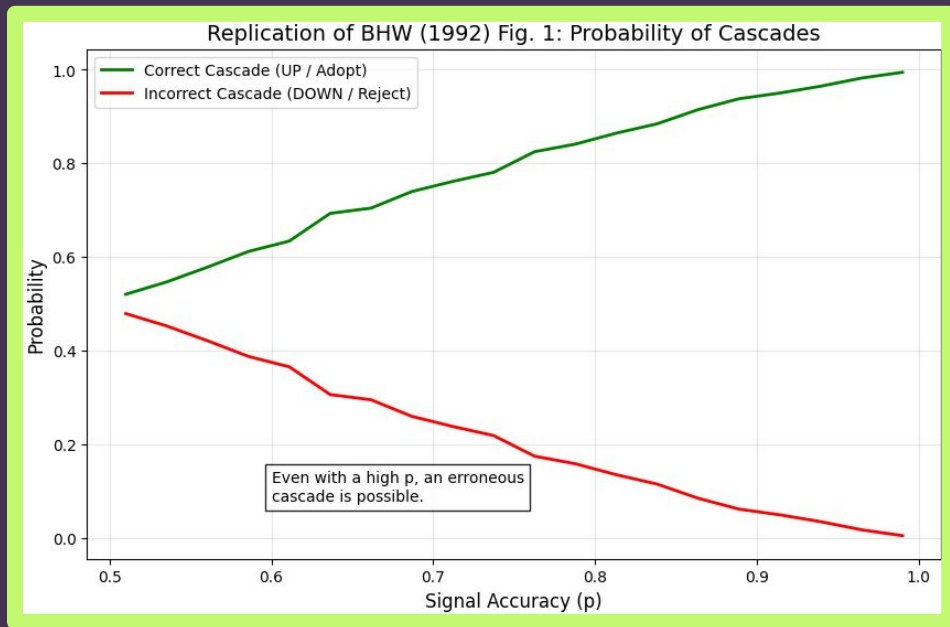
Goal: replicate BHW quantitative targets (**Fig. 1, Table 2, Result 4**)

Model: PAO, binary signals, tie-breaking (coin flip)

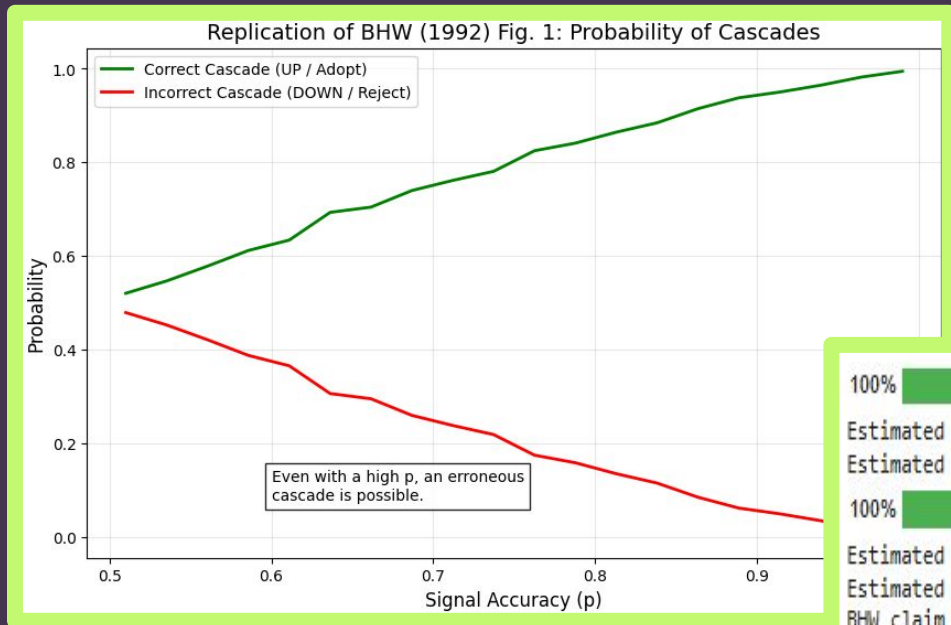
Method: Monte-Carlo simulation (N = 5000 runs per setting)

- Fig.1: $n_{\text{agents}} = 100$, p in $[0.51...0.99]$, $n_{\text{sims}} = 5000$
- Table2: $n_{\text{agents}} = 1000$, $p \in \{0.55, 0.65, 0.75, 0.85, 0.95\}$, $q \in \{0, 1/1000; 10/1000\}$, $n_{\text{sims}} = 10000$
- Result4: $n_{\text{sims}}=20000$, $\text{redraw_prob} = 0.1$

My Replication (Results)



My Replication (Results)



100% 20000/20000 [00:05<00:00, 3411.09it/s]

Estimated $P(\text{cascade reversal}) = 0.10095$

Estimated $P(\text{value actually changes}) = 0.04885$ (should be ≈ 0.05)

100% 20000/20000 [00:06<00:00, 3577.84it/s]

Estimated $P(\text{cascade reversal}) = 0.1009$ (95% CI: [0.0968, 0.1051])

Estimated $P(\text{value changes}) = 0.0488$ (95% CI: [0.0459, 0.0518]) (target ≈ 0.05)

BHW claim (Result 4): $P(\text{reversal}) > 0.0935$ while $P(\text{value changes})=0.05$

“Result 4: behavior may switch more often than fundamentals.”

My Replication (Results)

Replication of BHW (1992) Fig. 1: Probability of Cascades

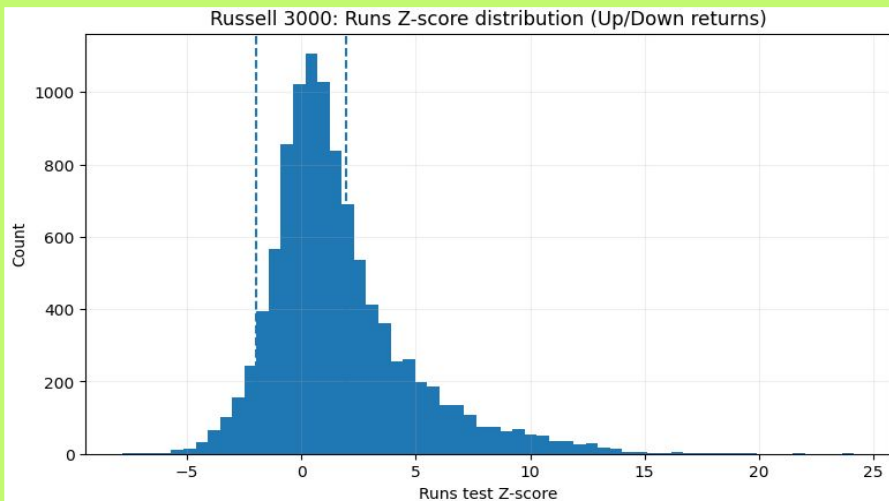


scenario	P(UP cascade @1000)			P(DOWN cascade @1000)			E[diff] ($\approx$ inferred H-L)		
	No public	Public(1/1000)	Public(10/1000)	No public	Public(1/1000)	Public(10/1000)	No public	Public(1/1000)	Public(10/1000)
p									
0.55	0.5593	0.5774	0.6290	0.4407	0.4220	0.3702	0.2372	0.4221	1.4560
0.65	0.6923	0.7203	0.8402	0.3077	0.2796	0.1594	0.7692	1.2132	4.2247
0.75	0.8027	0.8344	0.9562	0.1973	0.1654	0.0435	1.2108	1.8651	6.7275
0.85	0.8986	0.9255	0.9911	0.1014	0.0743	0.0089	1.5944	2.4266	8.8689
0.95	0.9731	0.9817	0.9996	0.0269	0.0183	0.0004	1.8924	2.8379	10.9774

Extra Outcomes (Market Analogue)

- **Stylized analogue (not structural test):** action proxy = sign of return (Up/Down)
- **Metric:** runs test Z-score $\rightarrow$ persistence ($Z < 0$) vs fragility/reversals ($Z > 0$)
- **Russell 3000 evidence:** market-wide distribution + relation to volatility (info intensity proxy)

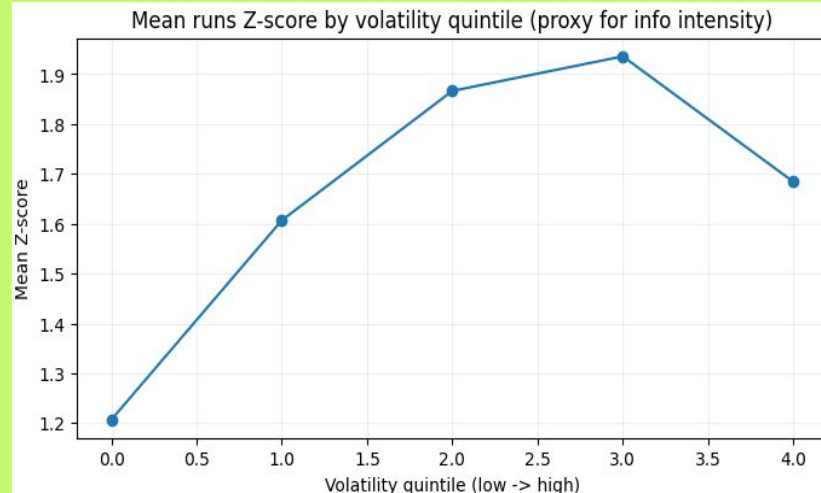
Extra Outcomes (Market Analogue)



Share $Z > +1.96$ (more reversals than RW): 0.348

Share $Z < -1.96$ (more persistence than RW): 0.062

Mean Z: 1.65994630920 Median Z: 0.98483228971



“More volatile stocks $\rightarrow$ higher Z (more reversals), consistent with fragility under information flow.”

“

Conclusion

Takeaways

Conformity can be **rational** (social learning)

Cascades are **error-prone** and **fragile**

Replication supports core predictions; market analogue shows “fragility” patterns

Limits: market proxy $\neq$ BHW micro-structure